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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Mole concept & Eudiometry

(English Medium)

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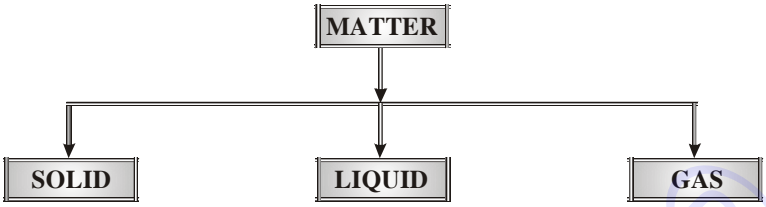
MOLE CONCEPT

1. MATTER : Matter is anything that has mass and occupies space.

Two ways of classifying matter :

I. Physical classification II. Chemical classification

1.1 Physical classification :

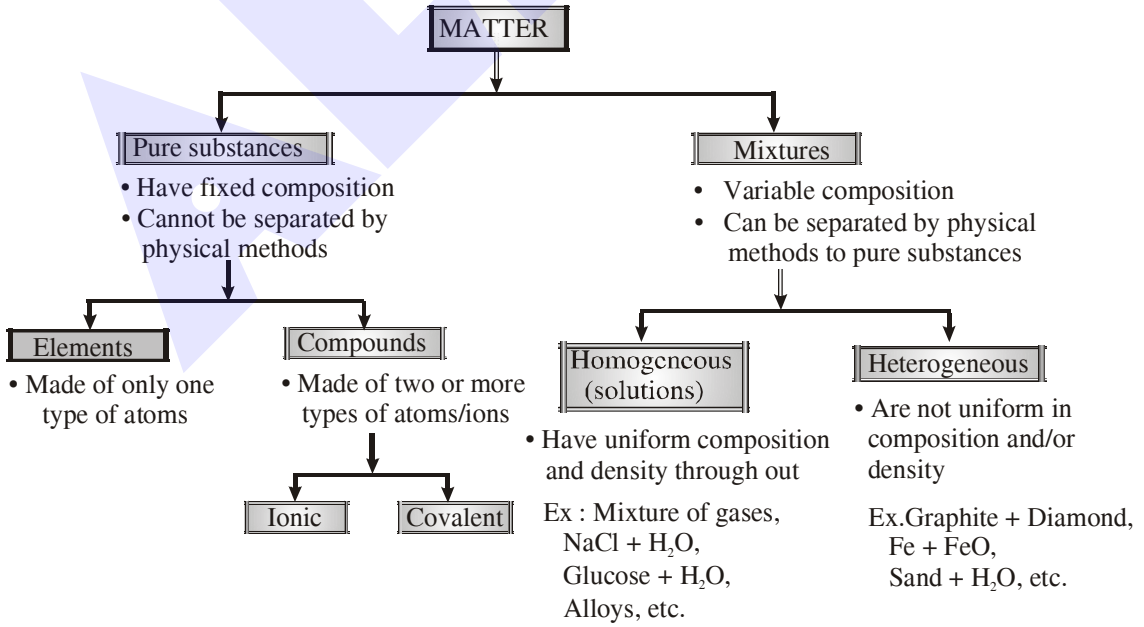


	SOLID	LIQUID	GAS
(i)	Particles held very closely packed in ordered manner.	Particles are less closely packed.	Particles are farthest apart
(ii)	No freedom of movement of particles	Particles can move around to some extent	Movement of particles is very easy and fast
(iii)	Definite shape and volume	Definite volume, indefinite shape	indefinite shape and volume
(iv)	Exists at low T and high P	Exists at intermediate P & T	Exists at high T and low P

Note : For same substance :

- Solid and Liquid co-exist at ***MELTING POINT***.
- Liquid and gas co-exist at ***BOILING POINT***.
- Solid and gas co-exist at ***SUBLIMATION POINT***.
- Solid, liquid and gas co-exist at ***TRIPLE POINT***.

1.2 Chemical classification :



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Note • **PHASE :** It is the state of matter uniform in density and composition.

• Homogeneous mixtures have single phase while heterogeneous mixtures are multi-phase.

Ex : NaCl + H_2O mixture has one phase

Ex : Graphite + Diamond mixture has 2 phases.

2. SOME SPECIFIC PROPERTIES OF SUBSTANCES

2.1 Deliquescence :

The property of certain compounds of taking up the moisture present in atmosphere and becoming wet when exposed, is known as deliquescence. These compounds are known as deliquescent. Sodium hydroxide, potassium hydroxide, anhydrous calcium chloride, anhydrous magnesium chloride, anhydrous feric chloride, etc., are the examples of deliquescent compounds.

2.2 Hygroscopicity :

Certain compounds combine with the moisture of atmosphere and are converted into hydroxides or hydrates. Such substances are called hygroscopic. Anhydrous copper sulphate, quick lime (CaO), anhydrous sodium carbonate, etc., are of hygroscopic nature.

2.3 Efflorescence :

The property of some crystalline substances of losing their water of crystallisation on exposure and becoming powdery on the surface is called efflorescence and such salts are known as efflorescent. The examples are : Ferrous sulphate ($FeSO_4.7H_2O$), sodium carbonate ($Na_2CO_3.10H_2O$), sodium sulphate ($Na_2SO_4.10H_2O$), potash alum [$K_2SO_4.Al_2(SO_4)_3.24H_2O$], etc.

2.4 Effervescence :

Effervescence is the escape of gas from an aquous solution and the foaming or fizzing that results from that release. Effervescence can be observed when opening a bottle of carbonated soft drinks. The visible bubbles are produced by the escape from solution of the dissolved gas (which itself is not visible while dissolved in the liquid.)

2.5 Malleability :

This property is shown by metals. When metallic solid is being beaten, it does not break but is converted into thin sheet. It is said to possess the property of malleability. Copper, gold, silver, aluminium, lead, etc., can be easily hammered into sheets. Gold is the most malleable metal.

2.6 Ductility :

The property of metal to be drawn into wires is termed ductility. Copper, silver, gold, aluminium, iron, etc., are ductile in nature. Platinum is the most ductile metal.

2.7 Brittleness :

The solid materials which break into small pieces on hammering are called brittle. The solids of non-metals and ionic solid are generally brittle in nature.

Ex: Ice, Diamond etc.

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3. THE LAW OF CHEMICAL COMBINATION

Antoine Lavoisier, John Dalton and other scientists formulated certain laws concerning the composition of matter and chemical reactions. These laws are known as the law of chemical combination.

3.1 Law of indestructibility of matter or conservation of Mass :



Antoine Lavosier
(1743-1794)

Antoine-Laurent de Lavosier, the "father of modern chemistry," was a French nobleman prominent in the histories of chemistry and biology. He named both oxygen and hydrogen and predicted silicon.

- This law was proposed by ***Lavoisier in 1789.***
- In all physical or chemical changes, the total mass of the system remains constant. In a physical or chemical change, mass is neither created nor destroyed. Thus, in a physical or chemical change.

Total mass of reactant reacted = Total mass of products formed

Ex. $H_2O(s) \longrightarrow H_2O(\ell)$

Above reaction shows the physical change and the mass of H_2O (s) = mass of $(H_2O)(\ell)$

In case the reacting materials are not completely consumed, the relationship will be.

Total masses of reactants = Total masses of product + masses of unreacted reactants

Note : In nuclear reactions (Mass + energy) is conserved, not the mass seperately.

Ex.1 ***When 4.2 g $NaHCO_3$ is added to a solution of CH_3COOH weighing 10.0 g, it is observed that 2.2 g CO_2 is released into atmosphere. The residue is found to weigh 12.0 g. Show that these observations are in agreement with the law of conservation of mass.***

Sol. $NaHCO_3 + CH_3COOH \longrightarrow CH_3COONa + H_2O + CO_2$

Initial total mass = 4.2 + 10 = 14.2

Final total mass = 12 + 2.2 = 14.2

Thus, during the course of reaction , law of conservation of mass is obeyed.

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3.2. Law of constant or definite proportion :

- This law was given by ***Joseph Louis Proust. in 1799***.
- Chemical compostion of a compound remains constant whether it is obtained by any method or any source.



Joseph Proust
(1754 - 1826)

Proust was born the son of anapothecary at Angers in north-west France. He studied in Paris.He lived in poverty for some years before being awarded a pension by Louis XVIII.

◆ **Example :**

In water (H_2O), Hydrogen and Oxygen combine in 1 : 8 mass ratio, the ratio remains constant whether it is tap water, river water or sea water or produced by any chemical reaction.

Note: This law is not applicable in case of isotopes.

Ex.2 ***1.80 g of a certain metal burnt in oxygen gave 3.0 g of its oxide. 1.50 g of the same metal heated in steam gave 2.50 g of its oxide. Show that these results illustrate the law of constant proportion.***

Sol. In the first sample of the oxide,

wt. of metal = 1.80 g , wt. of oxygen = (3.0 – 1.80) g = 1.2 g

$$\therefore \quad \frac{\text{wt.of metal}}{\text{wt.of oxygen}} = \frac{1.80\text{g}}{1.2\text{g}} = 1.5$$

In the second sample of the oxide,

wt. of metal = 1.50 g , wt. of oxygen = (2.50 – 1.50) g = 1 g

$$\therefore \quad \frac{\text{wt.of metal}}{\text{wt.of oxygen}} = \frac{1.50\text{g}}{1\text{g}} = 1.5$$

Thus, in both samples of the oxide the proportions of the weights of the metal and oxygen are fixed. Hence, the results follows the law of constant proportion.

3.3. The law of multiple proportion :

- This law was given by Dalton in 1803.
- If two elements combine to form more than one compound, then the different masses of one element which combine with a fixed mass of the other element, bear a simple ratio to one another.

Ex. Nitrogen and oxygen combine to form the following stable oxides –

N_2O	Nitrogen 28 parts	Oxygen 16 parts
N_2O_2	Nitrogen 28 parts	Oxygen 32 parts
N_2O_3	Nitrogen 28 parts	Oxygen 48 parts
N_2O_4	Nitrogen 28 parts	Oxygen 64 parts
N_2O_5	Nitrogen 28 parts	Oxygen 80 parts

The masses of oxygen which combine with same mass of nitrogen in the five compounds bear a ratio 16 : 32 : 48 : 64 : 80 or 1 : 2 : 3 : 4 : 5.

Note: This law is not applicable in case of isotopes.

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Ex.3 ***Carbon forms two oxides. One contains 27.27 % carbon & another contains 42.86% carbon. Show that the data illustrate the law of multiple proportion.***

Sol. The mass ratio of C : O in first oxide = 27.27 : 72.73 = 3 : 8

The mass ratio of C : O in second oxide = 42.86 : 57.14 = 3 : 4

Hence for each 3 gm of carbon the masses of oxygen combined is in 2 : 1 ratio.

Hence the data is according the law of multiple proportion.

3.4 Law of Gaseous volumes :

- This law was given by ***Gay-Lussac.*** in 1808.
- According to this law, gases react with each other in the simple ratio of their volumes and if products are also gaseous then they are also formed in simple ratio of volume provided that all volumes are measured at same temp. & pressure.

eg. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

1 vol. 3vol. 2vol.



Joseph Louis Gay Lussac (1778 - 1850)

Joseph Louis Gay-Lussac also ; 6 December 1778 – 9 May1850)was a French chemist and physicist. He is known mostly for two laws related to gases, and for his work on alcohol-water mixtures , which led to the degrees Gay-Lussac used to measure alcoholic beverages in many countries.

Ex.4 ***For the gaseous reaction , $H_2 + Cl_2 \longrightarrow 2HCl$. If 40 ml of hydrogen completely reacts with chlorine then find out the required volume of chlorine and volume of produced HCl ?***

Sol. According to Gay Lussac's Law :

$H_2 + Cl_2 \longrightarrow 2HCl$

$\because$ 1 ml of H_2 will react with 1 ml of Cl_2 and 2 ml of HCl will be produced.

$\therefore$ 40 ml of H_2 will react with 40 ml of Cl_2 and 80 ml of HCl will be produced.

required vol. of Cl_2 = 40 ml, produced vol. of HCl = 80 ml

3.5. Avogadro's law (1811) :

Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules.



Lorenzo Romano Amedeo Carlo Avogadro di Quareqa edi Carreto (1776–1856)

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4. DALTON'S ATOMIC THEORY

Ancient Indian and Greek philosophers have always wondered about the unknown and unseen form of matter. The idea of divisibility of matter was considered long back in India, around 500 BC. An Indian philosopher ***Maharishi Kanad***, postulated that if we go on dividing matter (padarth), we shall get smaller and smaller particles. Ultimately, a time will come when we shall come across the smallest particle beyond which further division will not be possible. He named these particles **Parmanu**. Another Indian philosopher, Pakudha Katyayama, elaborated this doctrine and said that these particles normally exist in a combined form which gives us various forms of matter. Around the same era, the Greek philosopher Democritus expressed the belief that all matter consists of very small, indivisible particles, which he named ***atomos*** (meaning uncuttable or indivisible).



John Dalton (1766 - 1844) , an Englishman, began teaching at a Quaker school when he was 12. His fascination with science included an intense interest in meterology (he kept careful daily weather records for 46 years), which led to an interest in the gases of the air and their ultimate components, atom. Dalton is best known for his atomic theory, in which he postulated that the fundamental differences among atoms are their masses. He was the first to prepare a table of relative atomicweight.

Although Democritus' ideal was not accepted by many of his contemporaries (notably Plato and Aristotle), some how it endured. Experimental evidence from early scientific investigations provided support for the notion of "atomism" and gradually gave rise to the modern definitions of elements and compounds. It was in ***1808, John Dalton***, formulated a precise definition of the indivisible building blocks of matter that we call atoms. Dalton's work marked the beginning of the modern era of chemistry. The hypotheses about the nature of matter on which Dalton's atomic theory is based can be summarized as follows :

(i) Matter consists of indivisible atoms.

(ii) All atoms of a given element are identical, having the same size, mass and chemical properties. The atoms of one element are different from the atoms of all other elements.

(iii) Compounds are formed when atoms of different elements combine in a fixed ratio.

(iv) A chemical reaction involves only the separation, combination or rearrangement of atoms. It does not result in their creation or destruction.

5. ATOMIC AND MOELCULAR MASSES :

The mass of an atom depends on the number of electrons, protons, and neutrons it contains. Knowledge of an atom's mass is important in laboratory work. But atoms are extremely small particles - even the smallest speck of dust that our unaided eyes can detect contains as many as 1×10^{16} atoms ! Clearly we cannot weigh a single atom, but it is possible to determine the mass of one atom relative to another experimentally. The first step is to assign a value to the mass of one atom of a given element so that it can be used as a standard.

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5.1 RELATIVE ATOMIC MASS :

Hydrogen, being lightest atom was arbitrarily assigned a mass of 1 (without any units) and other elements were assigned masses relative to it. However, the present system of atomic masses is based on carbon - 12 as the standard and has been agreed upon in 1961. Here, Carbon - 12 is one of the isotopes of carbon and can be represented as ^{12}C. In this system, ^{12}C is assigned a mass of exactly 12 atomic mass unit **(amu)** and masses of all other atoms are given relative to this standard.

Relative Atomic Mass is defined as the number which indicates how many times the mass of one atom of an element is heavier in comparison to 1/12th part of the mass of one atom of C-12.

$$\text{Relative atomic mass of an element} = \frac{\text{mass of one atom of an element}}{\frac{1}{12}[\text{mass of one C-12 atom}]}$$

$$= \frac{\text{Mass of one atom of an element}}{1\,\text{amu}}$$

5.1.1 ATOMIC MASS UNIT (a.m.u. or u) : The quantity $1/12^{th}$ mass of an atom of C^{12} is known as atomic mass unit.

Since mass of 1 atom of C - 12 $= 1.992648 \times 10^{-23}$ g

$\therefore$ $1/12^{th}$ part of the mass of 1 atom $= \frac{1.992648 \times 10^{-23}\,g}{12} = 1.67 \times 10^{-24}$ g $= \frac{1}{6.022 \times 10^{23}}$ g

It may be noted that the atomic masses as obtained above are the relative atomic masses and not the actual masses of the atoms. These masses on the atomic mass scale are expressed in terms of atomic mass units (abbreviated as amu). Today, 'amu' has been replaced by 'u' which is known as **unified mass**.

5.1.2 GRAM ATOMIC MASS OR MASS OF 1 G-ATOM :

When numerical value of atomic mass of an element is expressed in grams then the value becomes gram atomic mass or GAM.

gram atomic mass (GAM) = mass of 1**g-atom** = mass of 1 **mole atoms**

= mass of N_A atoms = mass of 6.022×10^{23} atoms.

Ex. GAM of oxygen= mass of 1 **g-atom** of oxygen = mass of 1 **mol atoms** of oxygen.

$$= \text{mass of } N_A \text{ atoms of oxygen} = \left(\frac{16}{N_A}g\right) \times N_A = 16\ g$$

Ex. Mass of one atom of Oxygen = 16 amu or $16 \times 1.66 \times 10^{-24}$ g

Mass of N_A atoms of Oxygen = $16 \times 1.66 \times 10^{-24} \times 6.022 \times 10^{-23}$ g = 16 g

Now see the table given below and understand the definition given before.

Element	R.A.M. (Relative Atomic Mass)	Atomic mass (mass of one atom)	Gram Atomic mass or weight
N	14	14 amu	14 gm
He	4	4 amu	4 gm
C	12	12 amu	12 gm

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5.1.3 AVERAGE ATOMIC MASS :

If an element exists in different isotopic forms having relative abundance $X_1\%$, $X_2\%$ ..... $X_n\%$, with relative atomic masses M_1, M_2 ..... M_n respectively then ,

Avg. Atomic mass of element $= \frac{X_1}{100}(M_1) + \frac{X_2}{100}(M_2) + ... + \frac{X_n}{100}(M_n) = \sum_{i=1 \text{ to } n} \frac{X_i}{100}(M_i)$

Ex.5 ***The atomic mass of an element is 50***

(i) Calculate the mass of one atom, in amu

(ii) Calculate the mass of 6.022×10^{23} atoms, in gm

(iii) Calculate the number of atoms in its 10 gm

(iv) What mass of the element contains 3.011×10^{20} atoms

Sol. (i) 50 amu

(ii) 50 gm

(iii) $\because$ 50 gm of element contains 6.022×10^{23} atoms

$\therefore$ 10 gm of element will contain $\frac{6.022 \times 10^{23}}{50} \times 10 = 1.2044 \times 10^{22}$ atoms

(iv) $\because$ 6.022×10^{23} atoms weighs 50 gm

$\therefore$ 3.011×10^{20} atoms weighs $\frac{50}{6.022 \times 10^{23}} \times 3.011 \times 10^{20} = 0.025$ gm

Ex.6 ***An element exist in nature in two isotopic forms : X^{30} (90%) and X^{32}(10%). What is the average atomic mass of element ?***

Sol. Av. atomic mass $= \frac{\Sigma(\%\text{abundance} \times \text{atomic mass})}{100} = \frac{90 \times 30 + 10 \times 32}{100} = 30.2$

5.2 RELATIVE MOLECULAR MASS :

The number which indicates how many times the mass of one molecule of a substance is heavier in comparison to 1/12th part of the mass of an atom of C–12.

OR

The molecular mass of a substance is the sum of atomic masses of all the atoms present in a molecule. It is obtained by multiplying the atomic mass of each element by the number of its atoms and adding them together.

Ex. molecular mass of oxygen (O_2) = 32

molecular mass of (O_3) = 48

molecular mass of HCl = 1 + 35.5 = 36.5

molecular mass of H_2SO_4 = 2 + 32 + 64 = 98

5.2.1 GRAM MOLECULAR MASS (MASS OF 1 G-MOLECULE) :

When numerical value of molecular mass of the substance is expressed in grams then the value becomes gram molecular mass or GMM.

gram molecular mass (GMM) = mass of 1 **g-molecule** = mass of 1 **mole molecules**

= mass of N_A molecules = mass of 6.022×10^{23} molecules

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Ex. GMM of H_2SO_4 = mass of 1 **g-molecule** of H_2SO_4

= mass of 1 **mole molecules** of H_2SO_4

= mass of N_A molecules of H_2SO_4

= $\left(\frac{98}{N_A}g\right)\times N_A = 98$ g

Ex. Molecular Mass of N_2 = 28 amu = $28 \times 1.66 \times 10^{-28}$ g

Mass of N_A molecules of N_2 = $28 \times 1.66 \times 10^{-24} \times 6.022 \times 10^{23}$ g = 28 g

5.2.2 AVERAGE MOLECULAR MASS OF NON-REACTING GAS MIXTURE :

$$M_{avg.} = \frac{\text{Total mass of mixture}}{\text{Total mole}} = \frac{\Sigma(\%\text{vol}\times\text{molecular mass})}{100} = \frac{100}{\Sigma\left(\frac{\%\text{mass}}{\text{molecular mass}}\right)}$$

Ex.7 ***The molecular mass of a compound is 75***

(i) Calculate the mass of 100 molecules, in amu.

(ii) Calculate the mass of 5000 molecules, in gm.

(iii) What is the mass of 6.022×10^{20} molecules, in gm

(iv) How many molecules are in its 2.5 mg

Sol. (i) mass of 1 molecules = 75 amu

$\therefore$ mass of 100 molecules = 7500 amu

(ii) Mass of 5000 molecules = 5000×75 amu

= $5000 \times 75 \times 1.66 \times 10^{-24} = 6.225 \times 10^{-19}$ gm

(iii) $\therefore$ 6.022×10^{23} molecules weighs 75 gm

$\because$ 6.022×10^{20} molecules weighs $\frac{75}{6.022\times10^{23}}\times 6.022\times10^{20} = 0.075$ gm

(iv) $\therefore$ 75 gm compound contains 6.022×10^{23} molecules

$\because$ 2.5×10^{-3} gm will contain $\frac{6.022\times10^{23}}{75}\times 2.5\times10^{-3} = 2.007 \times 10^{19}$ molecules.

Ex.8 ***A gaseous mixture contains 40% H_2 and 60% He, by volume. What is the average molecular mass of mixture ?***

Sol. $M_{av} = \frac{\Sigma(\%\text{ by vol.}\times\text{molecular mass})}{100} = \frac{40\times2+60\times4}{100} = 3.20$

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6 INTRODUCTION TO MOLE

Atoms and molecules are extremely small in size and their numbers in even a small amount of any substance is really very large. To handle such large numbers, a unit of similar magnitude is required. The 14th Geneva conference on weight and measures adopted **mole** as a ***seventh basic SI unit of the amount of a substance.*** Mole concept is essential tool for the fundamental study of chemical calculations. This concept is simple but its application requires a thorough practice. There are many ways of measuring the amount of substance, weight and volume being the most common, but basic unit of chemistry is the atom or a molecule and measuring the number of molecule is more important.

6.1 DEFINITION OF MOLE AND MOLAR MASS :

- A mole is the amount of a substance that contains as many entities (Atoms, Molecules, Ions or any other particles) as there are atoms in exactly 12 g of C-12 isotope.
- A mole of a substance contains Avogadro's number (6.022×10^{23}) of particles.
  The term mole, like a dozen or a gross, thus refers to a particular number of things. A dozen eggs equals 12 eggs, a gross of pencils equals 144 pencils, and a mole of ethanol equal 6.022×10^{23} ethanol molecules.
- The ***molar mass*** of a substance is the mass of one mole of the substance. Carbon-12 has a molar mass of exactly 12 g/mol , by definition.
- 1 g-atom = 1 mole atoms = N_A atoms
- 1 g-molecule = 1 mole molecules = N_A molecules
- 1 g-ion = 1 mole ions = N_A ions

6.2 Methods to calculate moles :

(i) If number of particles (molecules or atoms) is given then,

$$\text{mole} = \frac{\text{Given number of molecule / atom}}{N_A}$$

(ii) If mass is given then, number of mole $= \frac{\text{Given mass of substance (in gm)}}{\text{GAM / GMM}}$

(iii) If volume of gas is given then, mole

$$= \frac{\text{Volume of gas at STP}}{22.7\ \text{L}} = \frac{\text{Volume of gas at } 0^\circ\text{C and 1 atm}}{22.4\text{L}}$$

***** Standard molar volume is the volume occupied by 1 mole of an ideal gas gas at STP (Standard temperature and pressure which is 273.15 K & 1 bar respectively) , which is equal to 22.7 L.

* 1 mole of an ideal gas occupy 22.4 L at 0ºC and 1 atm.

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(iv) Under any condition of temperature and pressure, moles of gases may be calculated using **IDEAL GAS EQUATION :**

$$\boxed{PV = nRT}$$

where, R = Universal Gas Constant
= 0.082 L-atm/K-mol
= 8.314 J/K-mol
$\approx$ 2 cal/K-mol

Units of pressure and their relation:

1 atm = 76 cm Hg
= 760 mm Hg
= 760 torr (1 torr = 1 mm Hg)
= 1.01325×10^6 dyne/cm^2
= 1.01325×10^5 N/m^2 or Pa
= 1.01325 bar ($1 \text{ bar} = 10^5$ Pa)

1 bar = 75 cm Hg

Units of Volume and their relation:

1 ml = 1 cm^3 = 1 c.c.

1 Litre = 1000 ml = 1 dm^3

1 m^3 = 1000 L

Units of Temperature and their relation:

T = 273.15 + t

where, T = Absolute temperature (in Kelvin) and t = temperature in ^oC

(Normally , we take 273K in calculation)

(v) Sometimes gas is collected over water. In this case, the measured pressure is sum of pressure of gas and the vapour pressure of water (also called Aqueous Tension). In order to calculate moles of gas, the vapour pressure of water should be deducted from the measured pressure.

Ex.9 ***Calculate the number of g-molecules (mole of molecules) in the following : [$N_A = 6 \times 10^{23}$]***

(i) 3.2 gm CH_4
(ii) 70 gm nitrogen
(iii) 4.5×10^{24} molecules of ozone
(iv) 2.4×10^{21} atoms of hydrogen
(v) 11.2 L ideal gas at 0^oC and 1 atm
(vi) 4.54 ml SO_3 gas at STP
(vii) 8.21 L C_2H_6 gas at 400K and 2atm
(viii) 164.2 ml He gas at 27^oC and 570 torr

Sol. (i) 3.2 gram CH_4

$$\text{number of moles } (CH_4) = \frac{w}{M} = \frac{3.2}{16} = 0.2 \text{ moles}$$

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(ii) 70 gram N_2

Number of moles = $\frac{w}{M} = \frac{70}{28} = 2.5$

(iii) 4.5×10^{24} molecules of O_3

Number of moles = $\frac{\text{no. of molecules}}{N_A} = \frac{4.5 \times 10^{24}}{6 \times 10^{23}} = 7.5$

(iv) 2.4×10^{21} atoms of hydrogen

Number of gram molecules of H_2 = $\frac{\text{no. of molecules}}{N_A} = \frac{2.4 \times 10^{21}}{2 \times 6 \times 10^{23}} = 0.002$

(v) 11.2 litre ideal gas at 0ºC and 1 atm

Number of moles = $\frac{\text{Volume at } 0^\circ C \& 1 \text{ atm}}{22.4 \text{ litre}} = \frac{11.2}{22.4} = 0.5$

(vi) 4.54 ml SO_3 gas at STP

Number of moles = $\frac{V_{STP}(ml)}{22700 ml} = \frac{4.54}{22700} = 2 \times 10^{-4}$

(vii) 8.21 litre C_2H_6 at 400 K and 2 litre

$n = \frac{PV}{R.T} = \frac{2 \times 8.21}{0.0821 \times 400} = 0.5$

(viii) 164.2 ml He gas at 27ºC and 570 torr

$n = \frac{PV}{RT} = \left(\frac{570}{760} atm\right) \times \frac{164.2 \times 10^{-3} litre}{0.0821 \times 300} = 0.005$

Ex.10 ***Find no. of protons in 180 ml H_2O. Density of water = 1 gm/ml.***

Sol. Mass of water = density × volume = 180 g

Moles of water = $\frac{180}{18} = 10$

1 mol water has 10 mol protons

∴ 10 mol water has 100 mol protons = $100 \times 6.022 \times 10^{23}$ protons

= 6.022×10^{25} protons.

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Ex.11 ***What mass of $Na_2SO_4.7H_2O$ contains exactly 6.022×10^{22} atoms of oxygen ?***

Sol. Molar mass of $Na_2SO_4.7H_2O$ = 268 gm.

1 mole $Na_2SO_4.7H_2O$ has 11 mol O-atoms.

$\Rightarrow$ 11 N_A O – atoms are in 268 g $Na_2SO_4 . 7H_2O$

$$\Rightarrow 6.022 \times 10^{22} \text{ O – atoms are in } = \frac{268}{11 \times 6.022 \times 10^{23}} \times 6.022 \times 10^{22} \text{ g} = 2.44 \text{ g}$$

Ex.12 ***What is number of atoms and molecules in 112 L of $O_3(g)$ at 0°C and 1atm ?***

Sol. $$\text{Moles of molecules} = \frac{112}{22.4} = 5$$

Moles of atoms $= 5 \times 3 = 15$

No. of molecules = 5 N_A

No. of atoms = 15N_A.

7 DENSITY :

It is of two types.

I. Absolute density **II.** Relative density

7.1 For liquids and solids :

$$\text{Absolute density} = \frac{\text{mass}}{\text{volume}}$$

$$\textbf{Relative density or specific gravity} = \frac{\text{density of the substance}}{\text{density of water at 4°C (1gm ml}^{-1})}$$

7.2 For gases :

$$\text{Absolute density} = \frac{\text{mass}}{\text{volume}} = \frac{PM}{RT}$$

where P is pressure of gas, M = molar mass of gas, R is the gas constant, T is absolute temperature.

Vapour Density :

Vapour density is defined as the density of the gas with respect to hydrogen gas at the same temperature and pressure.

$$\text{Vapour density} = \frac{d_{gas}}{d_{H_2}} = \frac{PM_{gas}/RT}{PM_{H_2}/RT}$$

$$\text{V.D.} = \frac{M_{gas}}{M_{H_2}} = \frac{M_{gas}}{2} \Rightarrow \boxed{\mathbf{M_{gas} = 2 \times V.D.}}$$

Ex.13 ***A gaseous mixture of H_2 and NH_3 gas contains 68 mass % of NH_3. The vapour density of the mixture is –***

Sol. No. of moles of NH_3 in 100g mixture = $\frac{68}{17} = 4$

No. of moles of H_2 in 100g mixture = $\frac{32}{2} = 16$

$$M_{average} = \frac{\text{Total mass}}{\text{Total moles}} = \frac{100}{4+16} = 5$$

$$\text{V D} = \frac{5}{2} = 2.5$$

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8. STOICHIOMETRY

Stoichiometry is the calculation of amounts of reactants and products involved in a reaction. Stoichiometric calculations require a balanced chemical equation of the reaction.

A balanced chemical equation is one which contains an equal number of atoms of each element on both sides of equation.

8.1 SIGNIFICANCE OF STOICHIOMETRIC COEFFICIENTS :

Stoichiometric coefficients of chemical equation tells us about the ratio in which moles of reactants react and moles of products form.

Ex.	$2H_2(g)$ +	$O_2(g)$ ⟶	$2\ H_2O\ (g)$
1st interpretation	2 moles	1 mole	2 moles
2nd interpretation	$2\ N_A$ molecules	N_A molecules	$2\ N_A$ molecules
3rd interpretation	2 molecules	1 molecules	2 molecules

Ex.14 ***What mass of CaO is formed by heating 50 g $CaCO_3$ in air ?***

Sol. $CaCO_3\ (s) \longrightarrow CaO\ (s) + CO_2\ (g)$

50 gm

$= \frac{50}{100}\text{mol}$

$= \frac{1}{2}\text{mol}$ $\qquad \frac{1}{2}\text{mol} = \frac{1}{2}\times 56 = 28\text{ gm}$

Ex.15 ***If 1 mole of ethanol (C_2H_5OH) completely burns to form carbon dioxide and water, mass of carbon dioxide formed is about***

Sol. $C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$

1 mole $\qquad$ 2 mole

$\therefore$ mass of CO_2 formed $= 2 \times 44 = 88$ gm

Ex.16. ***What volume of CO_2 at 0°C and 1 atm is formed by heating 200 g $CaCO_3$?***

Sol. $CaCO_3\ (s) \longrightarrow CaO(s) + CO_2\ (g)$

200 gm

$= \frac{200}{100}\text{mol} = 2\text{ mol} \qquad 2\text{ mol}$

Volume of gas at 0°C and 1 atm = No. of moles × 22.4 L = 2 × 22.4 = 44.8 L.

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8.2 LIMITING REAGENT (L.R.) :

(i) The reactant which is completely consumed when a reaction goes to completion is called **Limiting Reactant or Limiting reagent.**

(ii) The reactant whose stoichiometrically taken amount is least, is limiting reactant.

$$\text{stoichiometrically taken amount} = \frac{\text{Given moles of reactant}}{\text{Stoichiometric coefficient of reactant in balance Reaction}}$$

(iii) When amounts of two or more than two reactants are given :

	aA	+	bB	⟶	cC	+	dD
Initial reacting mixture	n_A mol		n_B mol				
Stoichiometric amount	$\frac{n_A}{a}$		$\frac{n_B}{b}$				

If $\frac{n_A}{a} < \frac{n_B}{b}$ ⇒ A is limiting reagent.

If $\frac{n_A}{n_B} = \frac{a}{b}$ then reaction occurs to completion & no reactant is left at the end.

If $\frac{n_A}{a} > \frac{n_B}{b}$ ⇒ B is limiting reagent.

For calculation of moles of product, LR should be used.

Ex.17. ***28 gm Lithium is mixed with 48 gm O_2 to reacts according to the following reaction.***
$4Li + O_2 \longrightarrow 2Li_2O$ (Li = 7)
The mass of Li_2O formed is

Sol.

	$4Li$	+	O_2	⟶	$2Li_2O$
moles taken	$\frac{28}{7}$ = 4		$\frac{48}{32}$ = 1.5		
$\frac{\text{moles taken}}{\text{stoich.coeff.}}$	$\frac{4}{4} = 1$ (L.R.)		$\frac{1.5}{1} = 1.5$		

Moles of Li_2O formed $= \frac{2}{4} \times 4 = 2$

Mass of Li_2O formed = 2 × 30 = 60 gm

Ex.18 ***Calculate the mass of sucrose $C_{12}H_{22}O_{11}$ (s) produced by mixing 78 g of C(s), 11 g of H_2(g) & 67.2 litre of O_2 (g) at 0°C and 1 atm according to given reaction (unbalanced) ?***
$C(s) + H_2(g) + O_2(g) \rightarrow C_{12}H_{22}O_{11}(s)$

Sol. $12C(s) + 11\ H_2(g) + \frac{11}{2}O_2 \rightarrow C_{12}H_{22}O_{11}(s)$

Moles taken	$\frac{78}{12}$ = 6.5	$\frac{11}{2}$ = 5.5	$\frac{67.2}{22.4}$ = 3
$\frac{\text{moles taken}}{\text{stoich.coeff.}}$	$\frac{6.5}{12}$ = 0.54	$\frac{5.5}{11}$ = 0.5 (L.R.)	$\frac{3}{5.5}$ = 0.545

$\therefore$ Moles of $C_{12}H_{22}O_{11}$ formed $= \frac{5.5}{11} = 0.5$

Mass of sucrose obtained $= 0.5 \times 342 = 171$ grams.

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8.3 PROBLEMS BASED ON MIXTURE :

The composition of any mixture may be determined by reacting the mixture with some substance, by which either one or more component of mixture may react.

Ex.19 ***1.5 gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighing 1.46 gm. The reaction responsible for loss of weight is***

$$Fe_2O_3\,(s) \rightarrow Fe_3O_4\,(s) + O_2(g)$$

What is the percentage by mass of Fe_2O_3 in original sample.

Sol. $3Fe_2O_3\,(s) \rightarrow 2Fe_3O_4 + \frac{1}{2}O_2$

3×160 $\qquad$ $\frac{1}{2} \times 32$

$= 480$ gm $\qquad$ $= 16$ gm

loss of 16 gm $\rightarrow$ 480 gm Fe_2O_3

loss of 0.04 gm $\rightarrow 0.04 \times \frac{480}{16} = 1.2$ gm Fe_2O_3

% by mass $= \frac{1.2}{1.5} \times 100 = 80\%$

8.4 PERCENTAGE YIELD :

In general, when a reaction is carried out in the laboratory we do not obtain the theoretical amount of the product, in reality. The amount of the product that is actually obtained is called the **actual yield**. Knowing the actual yield and theorectical yield the percentage yield can be calculate as :

$$\boxed{\%\text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100}$$

The percentage yield of any product is always equal to the percentage extent of that reaction.

Ex.20 ***Aluminium reacts with sulphur to form aluminium sulphide. If 5.4 gm of aluminium reacts with 12.8gm sulphur & gives 12gm of aluminium sulphides, then the percent yield of the reaction is. (Al = 27, S = 32)***

Sol

	$2Al$	$+$	$3S$	$\longrightarrow$	Al_2S_3
Mole taken	$\frac{5.4}{27}$ gm		$\frac{12.8}{32}$ gm		
	$= 0.2$		$= 0.4$		
$\frac{\text{moles taken}}{\text{stoich.coeff.}}$	$\frac{0.2}{2}$		$\frac{0.4}{3}$		
	$= 0.1$ (L.R.)		$= 0.133$		

moles of Al_2S_3 formed $= \frac{1}{2} \times 0.2 = 0.1$

mass of Al_2S_3 formed $= 0.1 \times 150 = 15$ gm

But, from equation, only 12 gm Al_2S_3 is formed.

$$\%\text{ yield} = \frac{\text{actual yield}}{\text{theoritical yield}} \times 100 = \frac{12}{15} \times 100 = 80\,\%$$

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8.5 DEGREE OF DISSOCIATION, α :

It represents the mole of substance dissociated per mole of the substance taken.

$$A \rightarrow n \text{ particles; } \alpha = \frac{M_o - M}{(n-1).M}$$

where, n = number of product particles per particle of reactant

M_o = Molar mass of 'A'

M = Molar mass of final mixture

Dissociation decreases the average molar mass of system while association increases it.

Same formula is applicable for association, taking the correct value of 'n'.

Ex.21 ***For the reaction $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$***

Calculate degree of dissociation (α) if observed molar mass of mixture is 13.6

Sol. $M_0 = M_{NH_3} = 17$, M = 13.6 (given) , $n = \frac{4}{2} = 2$

$$\alpha = \frac{M_o - M}{(n-1).M} = \frac{17-13.6}{(2-1)\times 13.6} = 0.25$$

8.6 PERCENTAGE PURITY :

The percentage of a specified compound or element in an impure sample may be given as

$$\boxed{\%\text{ purity} = \frac{\text{Actual mass of pure substance}}{\text{Total mass of sample}} \times 100}$$

If impurity is unknown, it is always considered as inert (unreactive) material.

Ex.22 ***A chalk sample exactly requires 17.52 gram HCl for complete reaction with all $CaCO_3$ present in it. If the chalk sample is 72% pure, the mass of sample taken is***

Sol. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

Moles of HCl = $\frac{17.52}{36.5}$

Moles of $CaCO_3$ = $\frac{1}{2}\times\frac{17.52}{36.5}$

Weight of $CaCO_3$ required = $\frac{1}{2}\times\frac{17.52}{36.5}\times 100$

Mass of sample taken :

$= \frac{1}{2}\times\frac{17.52}{36.5}\times\frac{100\times 100}{72} = 33.33$ gm

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8.7 PROBLEMS RELATED WITH SEQUENTIAL REACTION :

When one of products formed in previous reaction is consumed in the next one.

Ex.23 ***How many grams H_2SO_4 can be obtained from 1320 gm PbS as per reaction sequence ?***

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

$$3SO_2 + 2HNO_3 + 2H_2O \longrightarrow 3H_2SO_4 + 2NO$$

[At. mass : Pb = 208, S = 32]

Sol. Moles of PbS = $\frac{1320}{240}$ = 5.5 mol

Moles of SO_2 = 5.5 mol = moles of H_2SO_4

Mass of H_2SO_4 = 5.5 × 98 = 539 gm

[When amount of only one reactant is given generally other is assumed in excess.]

Ex.24 ***Calcium phosphide Ca_3P_2 formed by reacting magnesium with excess calcium orthophosphate $Ca_3(PO_4)_2$, was hydrolysed by excess water. The evolved phosphine PH_3 was burnt in air to yield phosphrous pentoxide (P_2O_5). How many gram of magnesium metaphosphate would be obtain if 192 gram Mg were used (Atomic weight of Mg = 24, P = 31)***

$$Ca_3(PO_4)_2 + Mg \longrightarrow Ca_3P_2 + MgO$$

$$Ca_3P_2 + H_2O \longrightarrow Ca(OH)_2 + PH_3$$

$$PH_3 + O_2 \longrightarrow P_2O_5 + H_2O$$

$$MgO + P_2O_5 \longrightarrow Mg(PO_3)_2$$

magnesium metaphosphate.

Sol. Balanced chemical reaction :

$$Ca_3(PO_4)_2 + 8Mg \longrightarrow Ca_3P_2 + 8MgO$$

excess $\frac{192}{24} = 8$mole 1mol 8mole

$$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$$

1mole 2mole

$$2PH_3 + 4O_2 \longrightarrow P_2O_5 + 3H_2O$$

2mole 1mole

$$MgO + P_2O_5 \longrightarrow Mg(PO_3)_2$$

8mole 1mole (LR) 1 mole obtained

$W_{Mg(PO_3)_2}$ = 1 × 182 = 182 gm

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8.8 PROBLEM RELATED WITH PARALLEL REACTION :

When same two reactants form two or more products by independent reactions.

Ex.25 ***Carbon reacts with oxygen forming carbon monoxide and/or carbon dioxide depending an availablity of oxygen. Find moles of each product obtained when 160 gm oxygen reacts with (a) 12 g carbon (b) 120 g carbon (c) 72 g carbon.***

Sol. **(a)** $C + \frac{1}{2}O_2 \longrightarrow CO$ [initially use a reaction using lesser amount of oxygen]

t = 0	1mol	5mol		
t =∞	0	5–0.5	=	1mol
	(LR)	4.5mol		

Since CO & O_2 are left, CO_2 will also formed.

$$CO + \frac{1}{2}O_2 \longrightarrow CO_2$$

t = 0	1mol	4.5mol	0
t = ∞	0	4 mol	1 mol

At end, 1 mole CO_2 & no CO present

(b) $C + \frac{1}{2}O_2 \longrightarrow CO$

t = 0	10mol	5mol	0
t =∞	0	0	10mol

At end only 10 mol CO present.

(c) $C + \frac{1}{2}O_2 \longrightarrow CO$

t = 0	6mol	5mol	0
t =∞	0	2mol	6mol
	[LR]		

$$CO + \frac{1}{2}O_2 \longrightarrow CO_2$$

t = 0	6mol	2mol	0
t =∞	2mol	0 [LR]	4mol

At end [2mol CO + 4mol CO_2] left.

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Ex.26 ***25.4 gm of iodine and 14.2 gm of chlorine are made to react completely to yield mixture of ICl and ICl_3. Ratio of moles of ICl & ICl_3 formed is (Atomic mass : I = 127, Cl = 35.5)***

Sol.

$$I_2 + Cl_2 \longrightarrow ICl + ICl_3$$
$$0.1\,mol \quad 0.2\,mol \quad x\,mol \quad y\,mol$$

Applying conservations of I and Cl -atom

$0.2 = x+y \quad \Rightarrow \quad x = 0.1$

$0.4 = x+3y \quad \Rightarrow \quad y = 0.1$

$\therefore\ n_{ICl} : n_{ICl_3} = x : y = 1 : 1$

8.9 PRINCIPLE OF ATOM CONSERVATION (POAC)

POAC is nothing but the conservation of atoms of reactants and products involved in a chemical reaction. And if atoms are conserved, moles of atoms shall also be conserved. The principle is fruitful for the students when they don't get the idea of balanced chemical equation. In the problem using POAC we do not need to balance a reaction and we can even add two or more reactions. This principle can be understood by the following example.

Consider the decomposition of $KClO_3(s) \rightarrow KCl(s) + O_2(g)$ (unbalanced chemical reaction)

Apply the principle of atom conservation (POAC) for K atoms.

or moles of K atoms in $KClO_3$ = moles of K atoms in KCl

Now, since 1 molecule of $KClO_3$ contains 1 atom of K

Thus, moles of K atoms in $KClO_3$ = 1 × moles of $KClO_3$

and moles of K atoms in KCl = 1 × moles of KCl

$\therefore$ moles of $KClO_3$ = moles of KCl or $\dfrac{\text{wt. of } KClO_3 \text{ in g}}{\text{mol. wt. of } KClO_3} = \dfrac{\text{wt. of KCl in g}}{\text{mol. wt. of KCl}}$

- The above equation gives the mass-mass relationship between $KClO_3$ and KCl which is important in stoichiometric calculations.Again, applying the principle of atom conservation for O atoms,

moles of O in $KClO_3$ = 3 × moles of $KClO_3$

moles of O in O_2 = 2 × moles of O_2

$\therefore$ 3 × moles of $KClO_3$ = 2 × moles of O_2

or $$3 \times \frac{\text{wt. of } KClO_3}{\text{mol. wt. of } KClO_3} = 2 \times \frac{\text{vol. of } O_2 \text{ at 1atm and 0°C}}{\text{Molar vol. (22.4 lt)}}$$

- The above equations thus gives the mass-volume relationship of reactants and products.

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Ex.27 ***27.6 g K_2CO_3 was treated by a series of reagents so as to convert all of its carbon to $K_2Zn_3[Fe(CN)_6]_2$. Calculate the weight of the product. [mol. wt. of K_2CO_3 = 138 and mol. wt. of $K_2Zn_3[Fe(CN)_6]_2$ = 698]***

Sol. Here we have no knowledge about series of chemical reactions but we know about initial reactant and final product, accordingly.

$$K_2CO_3 \xrightarrow[\text{Steps}]{\text{Several}} K_2Zn_3[Fe(CN)_6]_2$$

Since C atoms are conserved, applying POAC for C atoms,

moles of C in K_2CO_3 = moles of C in $K_2Zn_3[Fe(CN)_6]_2$

$1 \times$ moles of $K_2CO_3 = 12 \times$ moles of $K_2Zn_3[Fe(CN)_6]_2$

$$\frac{\text{wt. of } K_2CO_3}{\text{mol. wt. of } K_2CO_3} = 12 \times \frac{\text{wt. of the product}}{\text{mol. wt. of product}}$$

wt. of $K_2Zn_3[Fe(CN)_6]_2 = \frac{27.6}{138} \times \frac{698}{12} = 11.63$ g

9. PERCENTAGE COMPOSITION OF AN ELEMENT IN A COMPOUND :

Percentage of perticular element present in a compound given as ,

Mass % of an element

$$= \frac{\text{mass of that element in the compound}}{\text{molar mass of the compound}} \times 100$$

$$= \frac{\text{no. of atoms of that element} \times \text{Atomic mass}}{\text{molar mass of the compound}} \times 100$$

Let us understand it by taking the example of water (H_2O). Since water contains hydrogen and oxygen, the percentage composition of both these elements can be calculated as follows :

$$\text{Mass \% of an element} = \frac{\text{mass of that element in the compound} \times 100}{\text{molar mass of the compound}}$$

Molar mass of water $= 18$ g

$$\text{Mass \% of hydrogen} = \frac{2 \times 1}{18} \times 100 = 11.11$$

$$\text{Mass \% of oxygen} = \frac{16}{18} \times 100 = 88.89$$

9.1 PERCENTAGE DETERMINATION OF ELEMENTS IN ORGANIC COMPOUNDS :

All these methods are applications of POAC

Do not remember the formulae, derive them using the concept, its easy.

(a) **Liebig's method :** (*for Carbon and hydrogen*)

$$\text{Organic Compound (w)} \xrightarrow[\text{CuO}]{\Delta} (w_1)CO_2 + H_2O(w_2)$$

$$\% \text{ of C} = \frac{w_1}{44} \times \frac{12}{w} \times 100$$

$$\% \text{ of H} = \frac{w_2}{18} \times \frac{2}{w} \times 100$$

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(b) **Duma's method :** (*for nitrogen*)

Organic Compound (w) $\xrightarrow[CuO]{\Delta}$ $N_2 \rightarrow$ (P, V, T given)

use PV = nRT to calculate moles of N_2, n.

$\therefore$ % of N = $\frac{n \times 28}{w} \times 100$

(c) Kjeldahl's method : (for nitrogen)

$$O.C.(w) \xrightarrow[conc.\ \Delta]{H_2SO_4} (NH_4)_2SO_4 \xrightarrow[\Delta]{NaOH} NH_3(w_1)$$

% of N = $\frac{14}{17} \times \frac{w_1}{w} \times 100$

(d) **Sulphur :**

O.C. (w) + $HNO_3 \rightarrow H_2SO_4 + BaCl_2 \rightarrow (w_1)\ BaSO_4$

$\Rightarrow$ % of S = $\frac{w_1}{233} \times \frac{1 \times 32}{w} \times 100$.

(e) **Phosphorus :**

O.C. (w) + $HNO_3 \rightarrow H_3PO_4$ + [NH_3 + magnesia mixture ammonium molybdate] $\rightarrow$

$MgNH_4PO_4 \xrightarrow{\Delta} Mg_2P_2O_7\ (w_1)$

% of P = $\frac{w_1}{222} \times \frac{2 \times 31}{w} \times 100$

(f) **Carius method :** (*Halogens*)

$$O.C.\ (w) \xrightarrow{Na} \xrightarrow{HNO_3 + AgNO_3} AgX(w_1)$$

If X is Cl then colour = curdy white
If X is Br then colour = dull yellow
If X is I then colour = bright yellow
Flourine can't be estimated by this

% of X = $\frac{w_1}{(M.\ weight\ of\ AgX)} \times \frac{1 \times (At.wt.of\ X)}{w} \times 100$

Ex.28 ***Which of the following compound have same % composition of carbon :***
HCHO, CH_3COOH, $C_6H_{12}O_6$, CH_3COCH_3

Sol. HCHO, CH_3COOH, $C_6H_{12}O_6$

% C(HCHO) = $\frac{1 \times 12}{30} \times 100 = 40\%$

% C(CH_3COOH) = $\frac{2 \times 12}{60} \times 100 = 40\%$

% C($C_6H_{12}O_6$) = $\frac{6 \times 12}{180} \times 100 = 40\%$

% C(CH_3COCH_3) = $\frac{3 \times 12}{58} \times 100 = 62.07\%$

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Ex.29 ***A sample of 0.5 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed by 2.45 gm of H_2SO_4. The residual acid required solution containing 1.2 gm. NaOH for neutralisation. Find the percentage composition of nitrogen in the compound ?***

Sol. $2\ NH_3 + H_2SO_4 \rightarrow (NH_4)_2\ SO_4$

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

Moles of H_2SO_4 taken $= \frac{2.45}{98} = 0.025$

Moles of NaOH $= \frac{1.2}{40} = 0.03$

$\therefore$ Moles of H_2SO_4 reacted with NaOH $= \frac{0.03}{2} = 0.015$

Remaining mol of $H_2SO_4 = 0.025 - 0.05 = 0.01$

mol of NH_3 evolved $= 0.01 \times 2 = 0.02$

% N in sample $= \frac{0.02 \times 14}{0.5} \times 100 = 56\%$

10. EMPIRICAL AND MOLECULAR FORMULA

We have just seen that knowing the molecular formula of the compound we can calculate percentage composition of the elements. Conversely if we know the percentage composition of the elements initially, we can calculate the relative number of atoms of each element in the molecules of the compound. This gives us the empirical formula of the compound. Further if the molecular mass is known then the molecular formula can be easily determined.

Thus, the **empirical** formula of a compound is a chemical formula showing the relative number of atoms in the simplest ratio, the **molecular formula** gives the actual number of atoms of each element in a molecule.

i.e. **Empirical formula :** Formula depicting constituent atoms in their simplest ratio.

Molecular formula : Formula depicting actual number of atoms in one molecule of the compound.

The molecular formula is generally an integral multiple of the empirical formula.

i.e. molecular formula = empirical formula × n

where $n = \frac{\text{molecular formula mass}}{\text{empirical formula mass}}$

Example :

Molecular Formula	H_2O_2 2 : 2	C_6H_6 6 : 6	C_2H_6 2 : 6	$C_2H_4O_2$ 2 : 4 : 2
Simplest ratio	1 : 1	1 : 1	1 : 3	1 : 2 : 1
Empirical Formula	H O	C H	CH_3	CH_2O

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10.1 DETERMINATION OF EMPIRICAL FORMULA :

Following steps are involved in determining the empirical formula of the compounds –

(i) First of all find the % by wt. of each element present in the compound.

(ii) The % by wt of each element is divided by its atomic weight. It gives atomic ratio of elements present in the compounds.

(iii) Atomic ratio of each element is divided by the minimum value of atomic ratio so as to get simplest ratio of atoms.

(iv) If the value of simplest atomic ratio is fractional then raise the value to the nearest whole number or multiply with suitable coefficient to convert it into nearest whole number

(v) Write the Empirical formula as we get the simplest ratio of atoms.

10.2 DETERMINATION OF MOLECULAR FORMULA :

(i) Find out the empirical formula mass by adding the atomic masses of all the atoms present in the empirical formula of compound.

(ii) Divide the molecular mass (determined experimentally by some suitable method) by the empirical formula mass and find out the value of n.

(iii) Multiply the empirical formula of the compound with 'n' so as to find out the molecular formula of the compound.

Ex.30. ***An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73. Molecular formula of compound is :-***

Sol. V.D. = 73 $\Rightarrow$ M = 2 × 73 = 146

$$C = 146 \times \frac{49.3}{100} = 71.978 \text{ g} = \frac{71.978}{12} \simeq 6 \text{ mole}$$

$$H = 146 \times \frac{6.84}{100} = 9.9864 \text{ g} = \frac{71.978}{12} \simeq 10 \text{ mole}$$

$$O = 146 \times \frac{100-(49.3+6.84)}{100} = \frac{64.86}{16} = 64.86 \text{ g} \approx 4 \text{ mol}$$

M.F. = $C_6H_{10}O_4$

Ex.31 ***The empirical formula of an organic compound containing carbon & hydrogen is CH_2. The mass of 1 litre of organic gas is exactly equal to mass of 1 litre N_2 therefore molecular formula of organic gas is.***

Sol. Empirical Mass of CH_2 = 12 + 2 = 14

$\because$ Mass of 1 litre of organic gas = Mass of 1 litre of N_2

Since V, P, T, n are same.

Therefore from $PV = \frac{m}{M}RT$ implies that molar mass should also be same.

$\therefore$ Molecular mass of organic compound will be 28 g

$$n = \frac{\text{Molecular mass}}{\text{Empirical mass}} = \frac{28}{14} = 2$$

So molecular formula = $2 \times CH_2 = C_2H_4$

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11. EUDIOMETRY :

Eudiometry or gas analysis involves the calculations based on gaseous reactions or the reactions in which at least two components are gaseous, in which the amounts of gases are represented by their volumes, measured at the same pressure and temperature. Some basic assumptions related with calculations are:

(i) Gay-Lussac's law of volume combination holds good. According to this law, the volumes of gaseous reactants reacted and the volumes of gaseous products formed, all measured at the same temperature and pressure, bear a simple ratio.

N_2 (g) + $3H_2$ (g) $\longrightarrow$ $2NH_3$ (g)
1 vol. 3 vol. 2 vol.

Problem may be solved directly is terms of volume, in place of mole. The stoichiometric coefficients of a balanced chemical reactions gives the ratio of volumes in which gaseous substances are reacting and products are formed, at same temperature and pressure.

(ii) The volumes of solids or liquids is considered to be negligible in comparison to the volume of gas. It is due to the fact that the volume occupied by any substance in gaseous state is even more than thousand times the volume occupied by the same substance in solid or liquid states.

$2H_2$ (g) + O_2 (g) $\longrightarrow$ $2H_2O$ (l)
2 mole 1 mole 2 mole
2 vol. 1 vol. 0 vol.

(iii) Air is considered as a mixture of oxygen and nitrogen gases only. It is due to the fact that about 99% volume of air is composed of oxygen and nitrogen gases only.

(iv) Nitrogen gas is considered as a non- reactive gas. It is due to the fact that nitrogen gas reacts only at very high temperature due to its very high thermal stability. Eudiometry is performed in an eudiometer tube and the tube can not withstand very high temperature. This is why, nitrogen gas can not participate in the reactions occurring in the eudiometer tube.

(v) The total volume of non-reacting gaseous mixture is equal to sum of partial volumes of the component gases **(Amagat's law)**.

$V = V_1 + V_2 +$.....

Partial volume of gas in a non-reacting gasesous mixture is its volume when the entire pressure of the mixture is supposed to be exerted only by that gas.

(vi) The volume of gases produced is often given by certain solvent which absorb contain gases.

Solvent	**Gases absorb**
KOH	CO_2, SO_2, Cl_2
Ammonical Cu_2Cl_2	CO
Turpentine oil	O_3
Alkaline pyrogallol	O_2
water	NH_3, HCl
$CuSO_4/CaCl_2$	H_2O

(vii) **EUDIOMETER**

An eudiometer is a laboratory device that measures the change in volume of a gas mixture following a physical or chemical change.

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Ex.32 ***10 ml of CO is mixed with 25 ml air (20% O_2 by volume). Find final volume (in ml) after complete combustion.***

Sol. $\overset{10ml}{CO} + \frac{1}{2}\overset{5ml}{O_2} \longrightarrow \underset{10ml}{CO_2}$

$V_f = V_{CO_2}$ + Volume of remaining air = 10 + 20 = 30 ml

Ex.33 ***A 3 L gas mixture of propane (C_3H_8) and butane (C_4H_{10}) on complete combustion at 25°C produced 10 L CO_2. Assuming constant P and T conditions what was volume of butane present in initial mixture ?***

Sol. $C_3H_8(g) + 5O_2 \longrightarrow 3CO_2(g) + 4H_2O(l)$

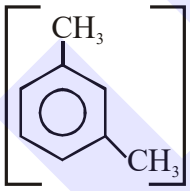
x L 3x L

$C_4H_{10}(g) + \frac{13}{2}O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$

(3-x) L 4(3 – x) L

from question $3x + 4(3 - x) = 10 \Rightarrow x = 2$

$\therefore$ Volume of butane , $C_4H_{10} = (3 - x) = 1$ L

Ex.34 ***100 ml gaseous meta Xylene***  ***undergoes combustion with excess of oxygen at room temperature and pressure. Volume contraction / expansion (in ml) during reaction is***

Sol.. $C_8H_{10}(g) + \frac{21}{2}O_2(g) \longrightarrow 8CO_2(g) + 5H_2O(l)$

100 ml $\frac{21}{2} \times 100 = 1050ml$ 800 ml 0

$\therefore$ Conctraction in volume = (100 + 1050) – 800 = 350 ml

Ex.35 ***An alkene upon combustion produces $CO_2(g)$ and $H_2O(g)$. In this combustion process if there is no volume change occurs then the no. of C atoms per molecule of alkene will be :***

Ans.(2)

$C_nH_{2n}(g) + \frac{3n}{2}O_2(g) \rightarrow nCO_2(g) + nH_2O(g)$

if there no volume changes i.e. $\Delta_{ng} = 0$

$(n + n) - \left(1 + \frac{3n}{2}\right) = 0 \Rightarrow n = 2$

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Ex.36 ***A gaseous hydrocarbon (C_xH_y) requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times of its volume of CO_2. Find out the value of (x + y)***

Ans. **(012)**

$$C_XH_y + (x + \frac{y}{4})\, O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O(l)$$

$$a \qquad a\left(x+\frac{y}{4}\right) \qquad ax$$

Given that : $a(x + y/4) = 6a$

and $ax = 4\,(a)$

$x = 4$ and $y = 8$ ...(2)

$\therefore\ x + y = 4 + 8 = 12$

Ex.37 ***On heating 60 ml mixture containing equal volume of chlorine gas and it's gaseous oxide, volume becomes 75 ml due complete decomposition of oxide. On treatment with KOH volume becomes 15 ml. What is the formula of oxide of chlorine ?***

Ans. **Cl_2O**

Sol. Let oxide of Cl is Cl_xO_y

So in 60 mL $\Rightarrow$ 30 mL Cl_xO_y and 30 mL Cl_2.

Now,

$$Cl_xO_y \longrightarrow \frac{x}{2}Cl_2 + \frac{y}{2}O_2$$

30mL

$$\frac{30.x}{2}\text{mL} \qquad \frac{30.y}{2}\text{mL}$$

Given :

$$75 = 30 + \frac{30x}{2} + \frac{30y}{2} \quad \Rightarrow \quad x + y = 3 \text{ .....(i)}$$

KOH absorbs Cl_2 and volume becomes 15 mL so,

$$(75 - 15) = V_{Cl_2} = 30 + \frac{30x}{2} \quad \Rightarrow \quad x = 2 \text{ and } y = 1$$

So the oxide : Cl_2O

Ex.38 ***5 L of A (g) & 3 L of B(g) measured at same T & P are mixed together which react as follows***

$$2A(g) + B(g) \rightarrow C\,(g)$$

What will be the total volume (in litre) after the completion of the reaction at same T & P.

Ans. **(3)**

Sol. $2A(g) + B(g) \longrightarrow C(g)$

5L 3L

L.R. is A

So, volume of C produced = $\frac{1}{2}\times 5 = 2.5$ L

and, volume of B reacted = $\frac{1}{2}\times 5 = 2.5$ L

So, volume fo B remained = 3 – 2.5 = 0.5 L

Hence, $V_{total} = V_C + V_B = 2.5 + 0.5 = 3$ L

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SELF STUDY

12. SIGNIFICANT FIGURES

Every experimental measurement has some amount of uncertainty associated with it. However, one would always like the results to be precise and accurate. Precision and accuracy are often referred to while we talk about the measurement.

Precision refers to the closeness of various measurements for the same quantity. However, **accuracy** is the agreement of a particular value to the true value of the result. For example, if the true value for a result is 2.00 g and a student 'A' takes two measurements and reports the results as 1.95 g and 1.93 g. These values are precise as they are close to each other but are not accurate. Another student repeats the experiment and obtains 1.94 g and 2.05 g as the results for two measurements. These observations are neither precise nor accurate. When a third student repeats these measurements and reports 2.01g and 1.99 g as the result. These values are both precise and accurate. This can be more clearly understood from the data given in Table

Data to Illustrate Precision and Accuracy

	Measurements/g			
	1	2	Average(g)	Remarks
Student A	1.95	1.93	1.940	Precise but not accurate
Student B	1.95	2.05	1.995	Neither precise nor accurate
Student C	2.01	1.99	2.000	Precise as well as accurate

The uncertainty in the experimental or the calculated values is indicated by mentioning the number of significant figures. **Significant figures** are meaningful digits which are known with certainty. The uncertainty is indicated by writing the certain digits and the last uncertain digit. Thus, if we write a result as 11.2 mL, we say the 11 is certain and 2 is uncertain and the uncertainty would be ± 1 in the last digit Unless otherwise stated, an uncertainty of ± 1 in the last digit is always understood.

There are certain rules for determining the number of significant figures. These are stated below:

(1) All non-zero digits are significant. For example in 285 cm, there are three significant figures and in 0.25 mL, there are two significant figures.

(2) Zeros preceding to first non-zero digit are not significant. Such zero indicates the position of decimal point.
Thus, 0.03 has one significant figure and 0.0052 has two significant figures.

(3) Zeros between two non-zero digits are significant. Thus, 2.005 has four significant figures.

(4) Zeros at the end or right of a number are significant provided they are on the right side of the decimal point. For example, 0.200 g has three significant figures.
But, if otherwise, the terminal zeros are not significant if there is no decimal point. For example, 100 has only one significant figure, but 100. has three significant figures and 100.0 has four significant figures.Such numbers are better represented in scientific notation. We can express the number 100 as 1×10^2 for one significant figure, 1.0×10^2 for two significant figures and 1.00×10^2 for three significant figures.

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(5) Counting numbers of objects, for example, 2 balls or 20 eggs, have infinite significant figures as these are exact numbers and can be represented by writing infinite number of zeros after placing a decimal i.e., 2 = 2.000000 or 20 = 20.000000

In numbers written in scientific notation, all digits are significant e.g., 4.01×10^2 has three significant figures, and 8.256×10^{-3} has four significant figures.

Addition and Subtraction of Significant Figures

The result cannot have more digits to the right of the decimal point than either of the original numbers.

$$\begin{array}{r} 12.11 \\ 18.0 \\ 1.012 \\ \hline 31.122 \end{array}$$

Here, 18.0 has only one digit after the decimal point and the result should be reported only up to one digit after the decimal point which is 31.1.

Multiplication and Division of Significant Figures

In these operations, the result must be reported with no more significant figures as are there in the measurement with the few significant figures.

$2.5 \times 1.25 = 3.125$

Since 2.5 has two significant figures, the result should not have more than two significant figures, thus, it is 3.1.

While limiting the result to the required number of significant figures as done in the above mathematical operation, one has to keep in mind the following points for rounding off the numbers

1. If the rightmost digit to be removed is more than 5, the preceding number is increased by one. for example, 1.386

   If we have to remove 6, we have to round it to 1.39

2. If the rightmost digit to be removed is less than 5, the preceding number is not changed.
   For example, 4.334 if 4 is to be removed, then the result is rounded upto 4.33.

3. If the rightmost digit to be removed is 5, then the preceding number is not changed if it is an even number but it is increased by one if it is an odd number. For example, if 6.35 is to be rounded by removing 5, we have to increase 3 to 4 giving 6.4 as the result. However, if 6.25 is to be rounded off it is rounded off to 6.2.

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SOLVED EXAMPLES

Ex.1 The weight of 2.01×10^{23} molecules of CO is- **[AIEEE 2002]**

(1) 9.3 g **(2) 7.2 g** **(3) 1.2 g** **(4) 3 g**

Ans. (1)

Sol. $\text{Mass} = \frac{2.01\times10^{23}}{6.02\times10^{23}} \times 28 = 9.3 \text{ gm}$

Ex.2 How many moles of magnesium phosphate, $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms? **[AIEEE 2006]**

(1) 3.125×10^{-2} **(2) 1.25×10^{-2}** **(3) 2.5×10^{-2}** **(4) 0.02**

Ans. (1)

Sol. $x \times 8 = 0.25 \Rightarrow x = 3.125 \times 10^{-2}$

Ex.3 The ratio of number of oxygen atoms (O) in 16.0 g ozone (O_3), 28.0 g carbon monoxide (CO) and 16.0 g oxygen (O_2) is :-
(Atomic mass : C = 12, O = 16 and Avogadro's constant $N_A = 6.0 \times 10^{23}$ mol^{-1})
[AIEEE 2012 (Online)]

(1) 3 : 1 : 1 (2) 1 : 1 : 2 (3) 3 : 1 : 2 (4) 1 : 1 : 1

Ans. (4)

$(\text{O-atoms})_{O_3} : (\text{O-atoms})_{CO} : (\text{O-atoms})_{O_2}$

$= \left(\frac{16}{48}\times N_A\right)\times3 : \left(\frac{28}{28}\times N_A\right)\times1 : \left(\frac{16}{32}\times N_A\right)\times2$

$= 1 : 1 : 1$

Ex.4 When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volume of hydrogen evolved is. **[JEE-1979]**

(A) 1 : 1 **(B) 1 : 2** **(C) 2 : 1** **(D) 9 : 4**

Ans. (A)

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$

$Zn + NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$

Ex.5 In the reaction **[AIEEE-2007]**

$2Al_{(s)} + 6HCl_{(aq)} \rightarrow 2Al^{3+}_{(aq)} + 6Cl^{-}_{(aq)} + 3H_2(g)$.

Select the correct information(s)

(1) 6L $HCl_{(aq)}$ is consumed for every 3L $H_2(g)$ produced

(2) 33.6 L $H_{2(g)}$ is produced regardless of temperature and pressure for every mole of Al that reacts.

(3) 67.2 L $H_{2(g)}$ at STP is produced for every mole of Al that reacts

(4) 11.2 L $H_{2(g)}$ at STP is produced for every mole of of $HCl_{(aq)}$ consumed

Ans. (4)

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Ex.6 If 0.50 mole of $BaCl_2$ is mixed with 0.20 mol of Na_3PO_4, the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is - [JEE-1981]

(A) 0.70 (B) 0.50 (C) 0.20 (D) 0.10

Ans. (D)

$3BaCl_2 + Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

0.5 mole 0.20 mole 0.1 mole

(L.R.)

Ex.7 What weight of AgCl will be precipitated when a solution containing 4.77 g of NaCl is added to a solution of 5.77 g of $AgNO_3$. (Ag = 108)

Ans. (4.879 gm)

$AgNO_3 + NaCl \rightarrow NaNO_3 + AgNCl$

5.77 gm 4.77 gm

0.034 mole (LR) 0.082 mole

wt. of AgCl = $0.034 \times 143.5 = 4.879$ gm

Ex.8 n-butane is product by the monobromination of ethane followed by the Wurtz reaction. Calculate the volume of ethane at NTP required to produce 55g n-butane, if the bromination takes place with 90% yield and the Wurtz reaction with 85% yield. [JEE-1984]

Ans. (55.50 litre)

(i) Monobromination of ethane

$$\underset{\text{Ethane}}{C_2H_6} \xrightarrow{Br_2} C_2H_5Br$$

(ii) Wurtz reaction :

$$2C_2H_5Br + 2Na + \xrightarrow[\text{ether}]{\text{Dry}} C_4H_{10} + 2NaBr$$

Molecular weight of

$C_4H_{10} = (12 \times 4) + (10 + 1) = 58$

$\therefore$ Amount of n-butane to be produced = $\frac{55}{58}$ mol = 0.948 mol

$\therefore$ Amount of C_2H_5Br required to obtain 0.948 mol

But the conversion is only 85%

Hence, the amount of C_2H_5Br required = $\frac{1.896}{85} \times 100 = 2.23$ mol

To obtain C_2H_5Br from C_2H_6, the same amount of C_2H_6 would be required. But the percent conversion of C_2H_6 to required = $\frac{2.23}{90} \times 100 = 2.478$mol.

Thus , required volume of ethane of NTP = $22400 \times 2.478 = 55507.2$ ml = 55.50 L.

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Ex.9 A solid mixture (5.0 g) consisting of lead nitrate and sodium nitrate was heated below 600°C unitll the weight of the residue was constant. If the loss in weight is 20% find the amount of lead nitrate and sodium nitrate in the mixture. [JEE-1990]

Ans. $Pb(NO_3)_2$ = 3.32 gm, $NaNO_3$ = 1.68 gm

Sol. $\underset{a\ gm}{Pb(NO_3)_2} \longrightarrow PbO + 2NO_2\uparrow + \frac{1}{2}O_2\uparrow$

$\underset{b\ gm}{NaNO_3} \longrightarrow NaNO_2 + \frac{1}{2}O_2\uparrow$

$\therefore$ a + b = 5 ..... (1)

The loss in weight for 5 g mixture = $5 \times \frac{28}{100} = 1.4g$

$\therefore$ Residue left = 5 – 1.4 = 3.6 g

The residue contain PbO + $NaNO_2$

$\because$ 331 g $Pb(NO_3)_2$ gives = 223g PbO

$\therefore$ ag $Pb(NO_3)_2$ gives = $\frac{223\times a}{332}$g PbO

Similarly ,

$\therefore$ 85 g $NaNO_3$ gives = 69 $NaNO_3$

$\therefore$ bg $NaNO_3$ gives = $\frac{69\times b}{85}$g $NaNO_3$

Solving equation, (1) and (2)

a = 3.32 g and b = 1.68 g

Ex.10 In an organic compound of molar mass 108 g mol^{-1} C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular formula can be : [AIEEE 2002]

(1) $C_6H_8N_2$ (2) $C_7H_{10}N$ (3) $C_5H_6N_3$ (4) $C_4H_{18}N_3$

Ans. (1)

Sol.

	C		H		N
	9	:	1	:	3.5
Mole	$\frac{9}{12}$	:	$\frac{1}{1}$	:	$\frac{3.5}{14}$
	0.75	:	1	:	0.25

C_3H_4N = emp. formula

mol. formula = $C_6H_8N_2$

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Ex.11 A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be [AIEEE 2012 (Online)]
(1) MCl_2 (2) MCl_4 (3) MCl_5 (4) MCl_3

Ans. 2

Let the formula of metal chloride is MCl_V.

Now, melecular weight , $2 \times 94.8 = A + V \times 35.5$ .....(1)

and $(2 \times 94.8) \times \frac{74.75}{100} = V \times 35$ .....(2)

From (2) ; $V = 3.99 \approx 4$

$\therefore$ formula is MCl_4

Ex.12 30 ml gaseous mixture of methane and ethylene in volume ratio X : Y requires 350 ml air containing 20% of O_2 by volume for complete combustion. If ratio of methane and ethylene changed to Y : X. What will be volume of air (in ml) required for complete reaction under similar condition of temperature and pressure.

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O\ (l)$

V_1ml $2V_1$ml V_1ml 0

$C_2H_4(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 2H_2O\ (l)$

V_2ml $3V_2$ml $2V_2$ml 0

For given data : $V_1 + V_2 = 30$

and $2V_1 + 3V_2 = 350 \times \frac{20}{100} = 70$

$\therefore\ V_1 = 20\ ,\ V_2 = 10$

For required data : $V_1 = 10$ and $V_2 = 20$

$\therefore$ Volume of O_2 required $= 2V_1 + 3V_2 = 80$ ml and volume of air required $= 80 = \frac{100}{20} = 400$ml

Ex.13 5 ml of gas containing only carbon and hydrogen was mixed with an excess of oxygen (30 ml) and the mixture exploded by means of an electric spark. After the explosion , the volume of the mixed gases remaining was 25 ml. On adding a concentrated solution of potassium hydroxide, the volume further diminished to 15 ml the residual gas being pure oxygen. All volumes have been reduced to NTP. Calculate the molecular formula of the hydrocarbon. [JEE-1979]

Ans. C_2H_4

Sol. Volume of oxygen taken = 30 ml

Volume of unused oxygen = 15 ml

Volume of O_2 used = Volume of O_2 added - volume of O_2 left = 30 – 15 = 15 ml.

and volume of CO_2 produced = 25 – 15 = 10 ml

General equation of the combustion of a hydrocarbon is as following lows.

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$$C_xH_y + \left(x + \frac{y}{4}\right) O_2 \rightarrow xCO_2 + \left(\frac{y}{2}\right) H_2O$$

5 ml $5\left(x + \frac{y}{2}\right)$ml 5x

$\therefore$ $5x = 10 \Rightarrow x = 2$

and $5\left(x + \frac{y}{2}\right) = 15$

$\therefore$ Molecular formula of hydrocarbon = C_2H_4

Ex.14 A 20.0 ml mixture of CO , CH_4 and He gases is exploded by an electric discharge at room temperature wth excess of oxygen. The volume contraction is found to be 13.0 cm^3. A further contraction of 14.0 cm^3 occurs when the residual gas is treated with KOH solution. Find out the composition of the gaseous mixture in terms of volume percentage. [JEE-1995]

Ans. (% CH_4 = 20, % CO = 50, % He = 30)

Sol. $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(l)$

'x' is the volume of CO and y is the volume of CH_4

Thus, $\frac{1}{2}x + 2y = 13$ (1)

$x + y = 14$ (2)

x = 10 and y = 4

Thus, % CH_4 = 20, % CO = 50, % He = 30

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